① $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

$AB = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} (3\cdot1 + 4\cdot0) & (3\cdot2 + 4\cdot-1) \\ (2\cdot1 + 1\cdot0) & (2\cdot2 + 1\cdot-1) \end{bmatrix}$

$= \boxed{\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}}$

$BA = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 3+4 & 4+2 \\ -2 & -1 \end{bmatrix} = \boxed{\begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}}$

$(A+B)^T = \left(\begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \right)^T = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}^T = \boxed{\begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}}$

$tr(A) = tr\left(\begin{bmatrix} 3 & 4 \\ 2 & -1 \end{bmatrix} \right) = 3 + -1 = \boxed{2}$

$tr(AB) = tr\left(\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \right) = 3 + 3 = \boxed{6}$

(2) $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

$M\vec{v} = \lambda\vec{v}$ $\quad M\vec{v} - \lambda\vec{v} = \vec{0}$

$M\vec{v} - \lambda I\vec{v} = \vec{0}$

$(M - \lambda I)\vec{v} = \vec{0}$

$\det(M - \lambda I) = 0$

$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}$ $\quad \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = 0$

Add up to B, multiply to C

$(2-\lambda)(2-\lambda) - 1 = 0$

$\lambda^2 - 4\lambda + 3 = 0$

$(\lambda - 1)(\lambda - 3) = 0$

$\boxed{\lambda = 1, 3}$

$\begin{bmatrix} 2-1 & -1 \\ -1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \rightarrow R_2 + 1(R_1) \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ $\quad \vec{V} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad x_1 - x_2 = 0$ $\quad x_1 = x_2$ $\quad \boxed{\vec{V} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$ $\quad \lambda = 1$

$\begin{bmatrix} 2-3 & -1 \\ -1 & 2-3 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \Rightarrow R_2 + 1(R_1) \begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-x_1 - x_2 = 0$ $\quad -x_1 = x_2$ $\quad \boxed{\vec{V} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}}$ $\quad \lambda = 3$

$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 2-1 \\ -1+2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ✓

$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ $\quad \begin{bmatrix} 2+1 \\ -1-2 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$ ✓

(3) trace(AB) = trace(BC)

trace(AB) = $(AB)_{11} + (AB)_{22} + \cdots + (AB_{nn})$ for an $n \times n$ AB

therefore trace(AB) = $a_{11} \cdot b_{11} + a_{12} b_{21} + \cdots a_{1n} b_{n1}$

$+ a_{21} b_{12} + a_{22} b_{22} + \cdots a_{2n} b_{n2}$

$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$

$+ \vdots$

$+ a_{n1} b_{1n} + a_{n2} b_{2n} + \cdots a_{nn} b_{nn}$

trace(BA) = $(BA)_{11} + (BA)_{22} + \cdots (BA_{nn})$

$= b_{11} a_{11} + b_{12} a_{21} + \cdots b_{1n} a_{n1}$

$+ b_{21} a_{12} + b_{22} a_{22} + b_{2n} a_{n2}$

$+ \vdots$

$+ b_{n1} a_{1n} + b_{n2} a_{2n} + \cdots b_{nn} a_{nn}$

When the traces are written in these forms,

Notice how the sum of the column in trace(AB)

is equal to the sum of the row in trace(BA) (BA_{kk})

therefore trace(AB) = trace(BA)

④ $C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$ has an inverse if the determinant is non 0

$\det(C) = 4 \cdot 6 - 7 \cdot 2 = 24 - 14 = \underline{10}$

therefore there is an inverse

$AA^{-1} = I$

$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad-bc & ab-ab \\ cd-cd & ad-bc \end{bmatrix} \Rightarrow \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} \frac{1}{ad-bc}$

$\frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = A^{-1}$ $\quad \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{7}{10} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$

Part 2 $P(X=k) = C \cdot \frac{3^k}{k!}$ $k = 0, 1, 2, \ldots$ Taylor Series

① a) We want to find the sum this distribution to, so we can divide by it to get the probability

$$C = \frac{1}{Z} \quad Z = \sum_{k=0}^{\infty} \frac{3^k}{k!} = e^3 \qquad C = \boxed{\frac{1}{e^3}} \qquad P(X) = \frac{3^x}{e^3 \cdot x!}$$

b) $E[X] = \sum_{x=0}^{\infty} x \cdot P(x)$ probability weighted outcome. $(x=k)$ summed

$$= \sum_{x=0}^{\infty} \frac{e^{-3} 3^x}{(x-1)!} \cdot x = 3e^{-3} \sum_{x=1}^{\infty} \frac{3^{x-1}}{(x-1)!} \qquad \text{let } k = x-1$$

$$= 3e^{-3} \sum_{k=0}^{\infty} \frac{3^k}{k!}$$

$$= 3e^{-3} e^3$$

$$= \boxed{3}$$

$$Var(X) = E[X^2] - (E[X])^2 \qquad E[X^2] = E[X(X-1)] + E[X]$$

$E[X] = 3$ $\quad E[X(X-1)] = \sum_{x=0}^{\infty} x(x-1) \frac{e^{-3} 3^x}{x!}$

$$= \sum_{x=2}^{\infty} \frac{e^{-3} 3^x}{(x-2)!} = e^{-3} 3^2 \sum_{x=2}^{\infty} \frac{3^{x-2}}{(x-2)!}$$

$k = x-2$

$$= 3^2 e^{-3} \sum_{k=0}^{\infty} \frac{3^k}{k!} = 3^2 e^{-3} (e^3) = 3^2$$

$E[X^2] = 3^2 + 3$

$Var[X] = 3^2 + 3 - 3^2 = \boxed{3}$

Part 2 (2) $P(A)=0.5$ $P(B)=0.3$ $P(A|B)=0.6$

a) $P(A \cap B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A|B) \cdot P(B) = P(A \cap B)$$

$$0.6 \cdot 0.3 = \boxed{0.18}$$

b) $P(B|A) = \frac{P(A|B)\,P(B)}{P(A)}$

$$\frac{0.6 \cdot 0.3}{0.5} = \frac{0.18}{0.5} = \boxed{0.9}$$

c) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= 0.5 + 0.3 - 0.18$$

$$0.8 - 0.18 = \boxed{0.62}$$

Pg 12 (3) a) $\theta = P(H)$ H, H, T, H, T H = 3 T = 2

$L(\theta) = \theta^H \cdot (1-\theta)^T$

$\ln(L(\theta)) = \ln(\theta^H \cdot (1-\theta)^T) = \ln(\theta^H) + \ln((1-\theta)^T) = H\ln\theta + T\ln(1-\theta)$

$\frac{d}{d\theta}\ln(L(\theta)) = \frac{d}{d\theta} H\ln\theta + T\ln(1-\theta) = \frac{d}{d\theta} H\ln\theta + \frac{d}{d\theta} T\ln(1-\theta)$ sum rule

$= \frac{H}{\theta} + \frac{T}{(1-\theta)} \cdot (-1)$ chain rule

$= \frac{H}{\theta} - \frac{T}{1-\theta} = 0$ $\frac{H}{\theta} = \frac{T}{1-\theta}$

$H(1-\theta) = T\theta$

$H - H\theta = T\theta$

$H = T\theta + H\theta$

$H = \theta(T+H)$

$H = \theta(5)$

$\theta = \frac{H}{5} = \boxed{\frac{3}{5}}$

b) $P(\theta) = \begin{cases} 2\theta, & 0 \le \theta \le 1 \\ 0 & \text{otherwise} \end{cases}$

$f(\theta | d_{n+1}) \propto L(\theta) \cdot P(\theta)$

$\theta^3 (1-\theta)^2 \cdot 2\theta$

$2\theta^4 (1-\theta)^2 \rightarrow \ln(2\theta^4 \cdot (1-\theta)^2) = \ln 2 + 4\ln\theta + 2\ln(1-\theta)$

$\rightarrow \frac{d}{d\theta}\ln 2 + \frac{d}{d\theta} 4\ln\theta + \frac{d}{d\theta} 2\ln(1-\theta) = \frac{4}{\theta} - \frac{2}{1-\theta}$

$\rightarrow 0 = \frac{4}{\theta} - \frac{2}{1-\theta}$ $\frac{2}{1-\theta} = \frac{4}{\theta}$ $2\theta = 4(1-\theta)$ $2\theta = 4 - 4\theta$

$6\theta = 4$

$\theta = \boxed{\frac{2}{3}}$

Par 2 (4) $x = 1, 3, 5, 7$

$$f(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$= \left(2\pi\sigma^2\right)^{-\frac{n}{2}} \exp\left(-\frac{\sum^{n}(x_i-\mu)^2}{2\sigma^2}\right)$$

$$\ln\left(L(\mu,\sigma^2)\right) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}$$

$$\frac{d}{d\mu}(\ln L) = 0 - 0 + \frac{2\sum_{i=1}^{n}(x_i-\mu)\cdot(-1)}{2\sigma^2}$$

$$\sigma^2 \cdot 0 = \frac{\sum_{i=1}^{n}(x_i-\mu)}{\sigma^2} \cdot \sigma^2$$

$$\sum_{i=1}^{n} x_i - \sum_{i=1}^{n}\mu = 0$$

$$\sum_{i=1}^{n} x^i - n\mu = 0 \qquad \sum_{i=1}^{n} x^i = n\mu \qquad \mu = \frac{\sum_{i=1}^{n} x_i}{n} \qquad \frac{1+3+5+7}{4} = \boxed{4} \quad \mu$$

$V = \sigma^2$

$$\frac{d}{dV}\ln L = 0 - \frac{n}{2V} + \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2V^2}$$

$$0 = -\frac{n}{2V} + \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2V^2}$$

$$\frac{n}{2V} = \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2V^2}$$

$$nV = \sum_{i=1}^{n}(x_i-\mu)^2 \qquad \sigma^2 = \frac{\sum_{i=1}^{n}(x_i-\mu)^2}{n}$$

$(1-4)^2 = 9$

$(3-4)^2 = 1$

$(5-4)^2 = 1$

$(7-4)^2 = 9$

$$\sigma^2 = \frac{9+1+1+9}{4} = \frac{20}{4} = \boxed{5}$$

Part 2 (5) $\mu \sim N(0,1)$

Likelihood - prior

$$L(\mu) \propto \exp\left(-\frac{(5-\mu)^2}{2(4)}\right) = \exp\left(-\frac{(5-\mu)^2}{8}\right)$$

$$P(\mu) \propto \exp\left(-\frac{(\mu-0)^2}{2(1)}\right) = \exp\left(-\frac{\mu^2}{2}\right)$$

$$f(\mu|x=5) \propto \exp\left(-\frac{(5-\mu)^2}{8}\right) \cdot \exp\left(-\frac{\mu^2}{2}\right)$$

$$\exp\left(-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}\right)$$

$$\ln f = -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}$$

$$\frac{d}{d\mu} \ln f = -\frac{2(5-\mu)(-1)}{8} - \frac{2\mu}{2}$$

$$\frac{5-\mu}{4} - \mu = 0$$

$$\frac{5-\mu}{4} = \mu \quad 5-\mu = 4\mu \quad 5 = 5\mu$$

$$\boxed{1 = \mu}$$

Part 2 (6) $Var(X) = E(X^2) - [E(X)]^2$

$$Var(X) = E[(X - E[X])^2]$$

$$= E[X^2 - 2XE[X] + E[X]^2]$$

$$= E[X^2] - 2E[X]E[X] + E[X]^2$$

$$= E[X^2] - 2E[X]^2 + E[X]^2$$

$$= \boxed{E[X^2] - E[X]^2}$$

Part 3 (1) $Var(u^T X) = u^T \Sigma u$ X is a random vector

$\mu = E[X]$ $\Sigma = E[(X-\mu)(X-\mu)^T]$ ← covariance matrix

u is a fixed unit vector

Let $Y = u^T X$ $E[Y] = E[u^T X] = u^T E[X] = u^T \mu$

$$Var(Y) = E[(Y - E[Y])^2]$$

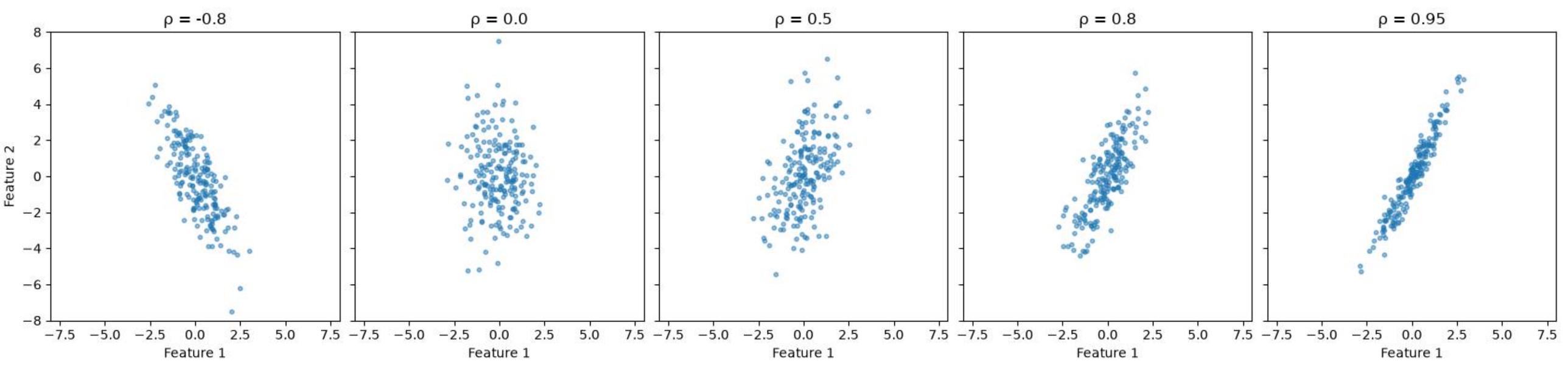
$$= E[(u^T X - u^T \mu)^2] = E[(u^T (X-\mu))^2]$$

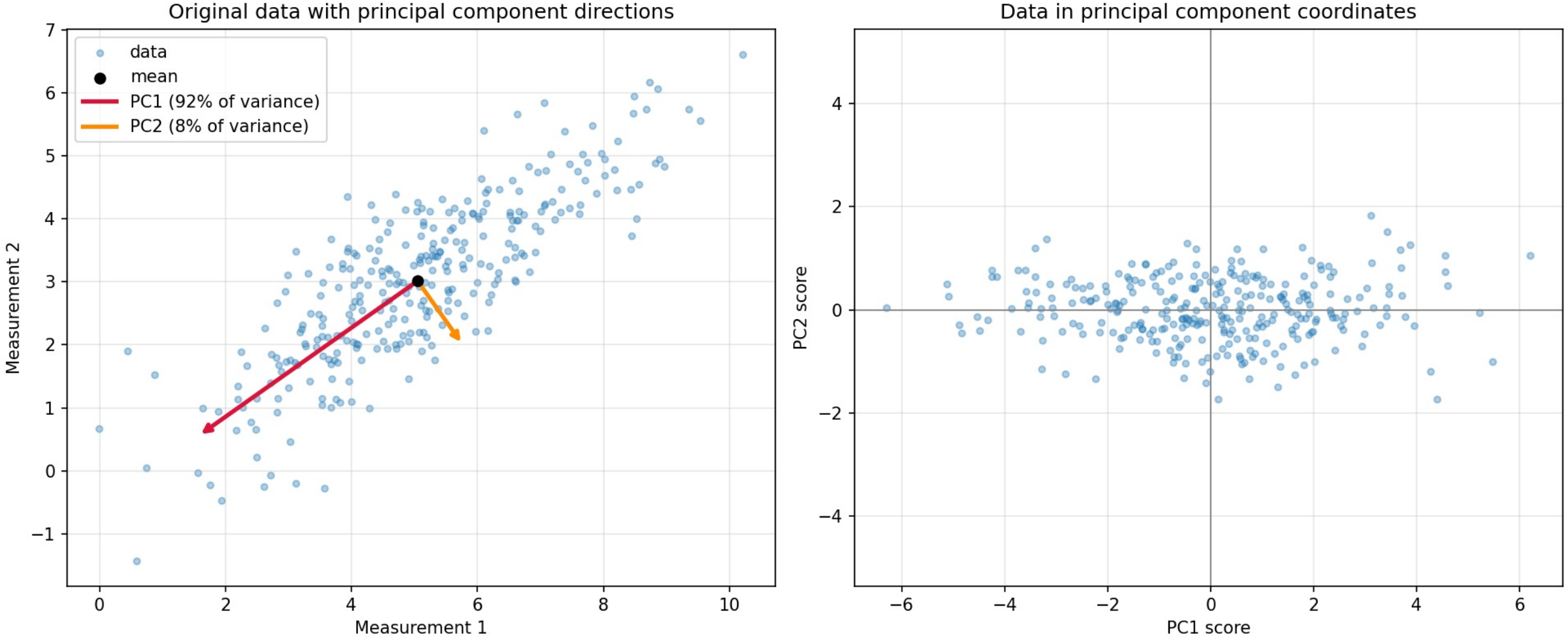
$$= E[(u^T (X-\mu))((X-\mu)^T u)]$$

$$= u^T E[(X-\mu)(X-\mu)^T] u$$

Σ ← covariance Matrix

$$\boxed{Var(u^T X) = u^T \Sigma u}$$


ρ = -0.8
ρ = 0.0
ρ = 0.5
ρ = 0.8
ρ = 0.95
Feature 2
Feature 1
8
6
4
2
0
−2
−4
−6
−8
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5


Original data with principal component directions
data
mean
PC1 (92% of variance)
PC2 (8% of variance)
Measurement 1
Measurement 2
Data in principal component coordinates
PC1 score
PC2 score